

Lecture 8. Probability and Distribution

Roughly speaking: the key insight for a sequence generative model (e.g. Large Language model) is to output the next word (given the previous output) with the highest probability

0. Basic concepts for Probability and Random variable.

1) Basic concepts

- ① Sample space Ω : The set of all possible outcome the experiment.
- ② Event space $\mathcal{A}$: The set of all subsets of Ω . The space of potential results of the experiment. $A \in \mathcal{A}$ with $A \subseteq \Omega$
- ③ Probability P : $P(A)$ the probability of $A \in \mathcal{A}$, which is a number in $[0, 1]$, measures the probability or degree of belief that an event $A \in \mathcal{A}$ will occur.
- ④ Target space: denote by T , associated with a mapping $X: \Omega \rightarrow T$ that maps event to a value in T
- ⑤ Random variable: the function X associated with T
- ⑥ Probability distribution: P_X , the law or distribution of random variable X .
- ⑦ Probability space $(\Omega, \mathcal{A}, P)$

(2) Discrete and Continuous probability: see section 6.2

1. Sum & product rule for probability.

Consider $\vec{x}$ and $\vec{y}$ are vectors of random variables

- ① $p(\vec{x}, \vec{y})$: denotes the joint distribution of $\vec{x}, \vec{y}$. (probability for a small neighbourhood around) the point $(\vec{x}, \vec{y})$.

(Borel σ -algebra)

② $p(\vec{y}|\vec{x})$: condition probability of $\vec{y}$ given $\vec{x}$ (given $\vec{x}$, how possible a $\vec{y}$ can be selected.)

(1) Product rule: $p(\vec{x}, \vec{y}) = p(\vec{y}|\vec{x}) \cdot p(\vec{x}) = p(\vec{x}|\vec{y}) \cdot p(\vec{y})$

(2) Sum rule (marginalization property).

$$p(\vec{x}) = \begin{cases} \sum_{\vec{y} \in Y} p(\vec{x}, \vec{y}), & \text{if } Y \text{ is a discrete set.} \\ \int_Y p(\vec{x}, \vec{y}) & \text{if } Y \text{ is a continuous set} \end{cases}$$

where Y stands for the states of the target space of random variable y .

Generally speaking, considering $\vec{X} = [x_1, \dots, x_D]^T$

$$p(x_i) = \int p(x_1, \dots, x_D) d\vec{x}_{\setminus i} \quad \left(\begin{array}{l} \text{integration over all dims} \\ \text{except for } x_i \end{array} \right)$$

called all except i

(3) Bayes' Theorem

$$p(\vec{x}|\vec{y}) = \frac{p(\vec{y}|\vec{x}) \cdot p(\vec{x})}{p(\vec{y})}$$

posterior likelihood prior evidence

① posterior: probability of $\vec{x}$ having observed $\vec{y}$

② Likelihood: The probability of $\vec{y}$ if we were to know the latent variable $\vec{x}$ (likelihood of $\vec{x}$ given $\vec{y}$ / probability of $\vec{y}$ given $\vec{x}$)

③ prior: ^{prior} knowledge of $\vec{x}$ before observing data $\vec{y}$

Note: Evidence / Marginal Likelihood.

$$\text{Discrete: } p(\vec{y}) := \sum_{\vec{x} \in X} p(\vec{y}|\vec{x}) \cdot p(\vec{x}) \quad \left. \vphantom{\sum_{\vec{x} \in X}} \right\} = E_{\vec{x}}[p(\vec{y}|\vec{x})]$$

$$\text{Continuous: } p(\vec{y}) = \int_{\vec{x} \in X} p(\vec{y}|\vec{x}) p(\vec{x}) d\vec{x} \quad \left. \vphantom{\int_{\vec{x} \in X}} \right\} = \underline{\underline{E_{\vec{x}}[p(\vec{y}|\vec{x})]}}$$

Exercise: $p(x, y)$ of two discrete random variable X, Y

	y_1	0.01	0.02	0.03	0.2
Y	y_2	0.05	0.1	0.05	0.27
	y_3	0.1	0.05	0.03	0.09
		x_1	x_2	x_3	x_4

compute condition distribution $P(y | X = x_i)$

$$P(X_4) = P(X_4, y_1) + P(X_4, y_2) + P(X_4, y_3) = 0.2 + 0.27 + 0.09 = 0.56$$

$$= \sum_{y \in Y} P(X_4, y)$$

$$P(y_1 | X = x_4) = \frac{P(y_1, x_4)}{P(x_4)} = \frac{0.2}{0.56}$$

$$P(y_2 | X = x_4) = \frac{P(y_2, x_4)}{P(x_4)} = \frac{0.27}{0.56}$$

$$P(y_3 | X = x_4) = \frac{P(y_3, x_4)}{P(x_4)} = \frac{0.09}{0.56}$$

2. Expectation

1) Expected value:

Def: The expected value of a function: $g: \mathbb{R}^X \mapsto \mathbb{R}$ of a random variable $X \sim p(x)$ is

$$E_X[g(x)] = \int_{x \in X} g(x) \cdot p(x) dx \quad (\text{continuous case})$$

$$\text{or } E_X[g(x)] = \sum_{x \in X} g(x) p(x)$$

Note: Multivariable: $X = [x_1, \dots, x_D]^T$, the expected value (element-wise)

$$E_X[g(\vec{x})] = \begin{bmatrix} E_{x_1}[g(x_1)] \\ \vdots \\ E_{x_D}[g(x_D)] \end{bmatrix} \in \mathbb{R}^D, \text{ with } E_{x_d} := d=1, \dots, D.$$

taking the expectation w.r.t. x_d , i.e. d -th element of $\vec{x}$

3, Mean.

For $\vec{X} = [X_1, \dots, X_D] \in \mathbb{R}^D$

$$E_{\vec{X}}[\vec{X}] = \begin{bmatrix} E_{X_1}(\vec{X}_1) \\ \vdots \\ E_{X_D}(\vec{X}_D) \end{bmatrix} \in \mathbb{R}^D, \text{ where.}$$

① $E_{X_d}(X_d) = \int_{\mathcal{X}} x_d p(x_d) dx_d$, if X is a continuous random variable.

② $E_{X_d}(X_d) = \sum_{x_i \in \mathcal{X}} x_i p(X_d = x_i)$, if X is a discrete random variable.

with $d = 1, \dots, D$

4, Linear of Expectation.

Let $f(\vec{x}) = a g(\vec{x}) + b h(\vec{x})$, for $a, b \in \mathbb{R}$, and $\vec{X} \in \mathbb{R}^D$

$$\begin{aligned} E_{\vec{X}}[f(\vec{x})] &= \int_{\mathcal{X}} f(\vec{x}) \cdot p(\vec{x}) d\vec{x} \\ &= \int_{\mathcal{X}} (a g(\vec{x}) + b h(\vec{x})) p(\vec{x}) d\vec{x} \\ &= \underbrace{a \int_{\mathcal{X}} g(\vec{x}) p(\vec{x}) d\vec{x}} + \underbrace{b \int_{\mathcal{X}} h(\vec{x}) p(\vec{x}) d\vec{x}} \\ &= a E_{\vec{X}}[g(\vec{x})] + b E_{\vec{X}}[h(\vec{x})] \end{aligned}$$

if $\vec{x} \in \mathcal{X}$ (discrete case).

$$\underline{E_{\vec{X}}[f(\vec{x})] = a E_{\vec{X}}[g(\vec{x})] + b E_{\vec{X}}[h(\vec{x})]}$$

Mean or Expected value describes what could be the outcome of the random variable, in average. Meanwhile, Variance shows how the actual outcome deviate from the mean.

5. Variance.

Def: The variance of random variable X with states $\vec{x} \in \mathbb{R}^D$ and the mean $\vec{\mu} \in \mathbb{R}^D$ is

$$\begin{aligned} V_X[\vec{x}] &:= \underline{E_X[(\vec{x} - \vec{\mu})(\vec{x} - \vec{\mu})^T]} \\ &= E_X[\vec{x}\vec{x}^T - \vec{x}\vec{\mu}^T - \vec{\mu}\vec{x}^T + \vec{\mu}\vec{\mu}^T] \\ &= E_X[\vec{x}\vec{x}^T] - E_X[\vec{x}\vec{\mu}^T] - E_X[\vec{\mu}\vec{x}^T] + E_X[\vec{\mu}\vec{\mu}^T] \\ &= E_X[\vec{x}\vec{x}^T] - E_X[\vec{x}]\vec{\mu}^T - \vec{\mu}E_X[\vec{x}^T] + E_X[\vec{x}]\vec{\mu}^T \\ &= E_X[\vec{x}\vec{x}^T] - E_X[\vec{x}]E_X[\vec{x}^T] - \cancel{E_X[\vec{x}]E_X[\vec{x}^T]} + \cancel{E_X[\vec{x}]E_X[\vec{x}^T]} \\ &\checkmark = \boxed{E_X[\vec{x}\vec{x}^T] - E_X[\vec{x}]E_X[\vec{x}^T]} \end{aligned}$$

$\vec{\mu} = E_X[\vec{x}]$
 $\vec{\mu}^T = E_X[\vec{x}^T]$

Let $Cov_X[X_i, X_j] = E_{X_i, X_j}[X_i X_j] - E_{X_i}[X_i]E_{X_j}[X_j]$

$$\checkmark V_X[\vec{x}] = \begin{bmatrix} Cov[X_1, X_1] & Cov[X_1, X_2] & \dots & Cov[X_1, X_D] \\ Cov[X_2, X_1] & Cov[X_2, X_2] & \dots & Cov[X_2, X_D] \\ \vdots & \vdots & \ddots & \vdots \\ Cov[X_D, X_1] & Cov[X_D, X_2] & \dots & Cov[X_D, X_D] \end{bmatrix} \Rightarrow \text{Covariance matrix}$$

Note: For a multivariate random variable, the variance describe the relation between individual dimensions of the random variable

→ What if One would like to describe how dependent random variables are to one another?

6. Covariance

① Def: Consider two multivariate random variable X and Y with states $\vec{x} \in \mathbb{R}^D$ and $\vec{y} \in \mathbb{R}^E$, respectively, the covariance between X and Y is defined as

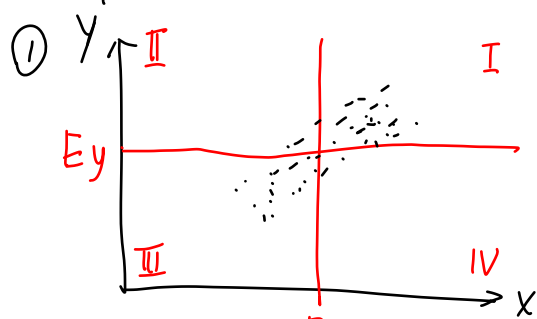
$$Cov[\vec{x}, \vec{y}] = E[\vec{x}\vec{y}^T] - E[\vec{x}]E[\vec{y}]^T = Cov[\vec{y}, \vec{x}]^T \in \mathbb{R}^{D \times E}$$

② Correlation: The correlation between two random variables X and Y , is

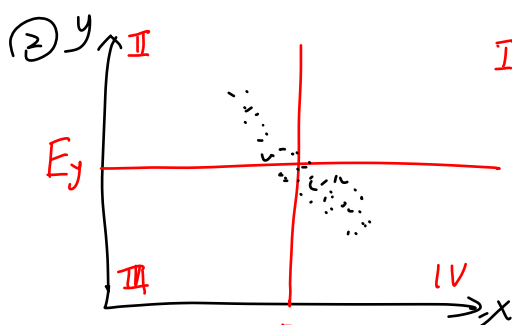
$$corr[x, y] := \frac{Cov[x, y]}{\sqrt{V[x]V[y]}} \in [-1, 1] \begin{cases} > 0, \text{ positive correlation} \\ < 0, \text{ negative correlation} \end{cases}$$

Note: Both covariance and correlation characterize how two random variables are related.

Graphical illustration.



$$\textcircled{1} E[(x - E_x)(Y - E_Y)] > 0$$



$$\textcircled{2} E[(x - E_x)(Y - E_Y)] < 0$$

Region I: $x > E_x, Y > E_Y, (x - E[x])(y - E[Y]) > 0$

Region II: $x < E_x, Y > E_Y, (x - E[x])(y - E[Y]) < 0$

Region III: $x < E_x, Y < E_Y, (x - E_x)(y - E_Y) > 0$

Region IV: $x > E_x, Y < E_Y, (x - E_x)(Y - E_Y) < 0$

7. Empirical Means and Covariances. ✓

Def: Consider a set of data $\{\vec{x}_i \in \mathbb{R}^D, i=1, \dots, N\}$

empirical mean: $\bar{x} := \frac{1}{N} \sum_{i=1}^N \vec{x}_i \in \mathbb{R}^D$

empirical covariance: $\Sigma := \frac{1}{N} \sum (\vec{x}_i - \bar{x})(\vec{x}_i - \bar{x})^T \in \mathbb{R}^{D \times D}$

$$\hookrightarrow \begin{cases} V_x[\vec{x}] = E_x[(x - \mu)(x - \mu)^T] : \stackrel{1-D}{=} E[(x - \mu)^2] \end{cases}$$

$$\stackrel{1-D}{=} V_x[x] = \frac{E_x(x^2) - (E_x[x])^2}{\frac{1}{N} \sum_{i=1}^N x_i^2 - (\frac{1}{N} \sum_{i=1}^N x_i)^2}$$

8. Basic Rules for Expectation and Variance

Consider Two random variable X, Y , with state $\vec{x}, \vec{y} \in \mathbb{R}^D$, Then

$$(1) E[\vec{x} \pm \vec{y}] = E[\vec{x}] \pm E[\vec{y}].$$

$$\text{Proof: (Discrete case): } E[X + Y] = \sum_{i,j} (\vec{x}_i + \vec{y}_j) P(X = \vec{x}_i, Y = \vec{y}_j)$$

$$= \sum_{i,j} \vec{x}_i P(X = \vec{x}_i, Y = \vec{y}_j) + \sum_{i,j} \vec{y}_j P(X = \vec{x}_i, Y = \vec{y}_j)$$

$$= \sum_i \vec{x}_i \underbrace{\sum_j P(X = \vec{x}_i, Y = \vec{y}_j)}_{P(X = \vec{x}_i)} + \sum_j \vec{y}_j \underbrace{\sum_i P(X = \vec{x}_i, Y = \vec{y}_j)}_{P(Y = \vec{y}_j)}$$

$$= \sum_i x_i P(X = x_i) + \sum_j y_j P(Y = y_j) = \underline{E[X]} + \underline{E[Y]}$$

$$\underline{E_x[\vec{x}]}$$

$$\textcircled{2} \quad V[X+Y] = V[X] + V[Y] + \text{Cov}[X, Y] + \text{Cov}[Y, X].$$

$$V[X+Y] = E[(X+Y) - E[X+Y]]^2$$

$$= E[(X - E[X]) + (Y - E[Y])]^2$$

$$= E[(X - E[X]) + (Y - E[Y])]^2$$

$$= (X - E[X])^2 + 2(X - E[X])(Y - E[Y]) + (Y - E[Y])^2$$

$$\Rightarrow E[(X - E[X])^2] + 2E[(X - E[X])(Y - E[Y])] + E[(Y - E[Y])^2]$$

$V[X] \qquad \text{Cov}[X, Y] \qquad V[Y]$

$$\Rightarrow \text{Let } \underline{\vec{y}} = \underline{A}\vec{x} + \underline{\vec{b}}, \quad \underline{\Sigma} := V_x[\vec{x}], \quad \underline{\vec{b}} \in \mathbb{R}^D, \text{ constant}$$

$$(3) E_Y[\vec{y}] = E_x[\underline{A}\vec{x} + \underline{\vec{b}}] = \underline{A}E_x[\vec{x}] + \underline{\vec{b}}$$

$$V_Y[\vec{y}] = V_x[\underline{A}\vec{x} + \underline{\vec{b}}] \stackrel{\textcircled{1}}{=} V_x[\underline{A}\vec{x}] \stackrel{\textcircled{2}}{=} \underline{A}V_x[\vec{x}]\underline{A}^T = \underline{A}\underline{\Sigma}\underline{A}^T$$

$$V_x(\underline{\vec{b}}) = E[\underline{\vec{b}}\underline{\vec{b}}^T] - E[\underline{\vec{b}}] \cdot E[\underline{\vec{b}}]^T = \underline{\vec{b}}\underline{\vec{b}}^T - \underline{\vec{b}}\underline{\vec{b}}^T = \underline{\vec{0}}$$

$$\text{Cov}(\vec{x}, \underline{\vec{b}}) = E[\vec{x}\underline{\vec{b}}^T] - E[\vec{x}]E[\underline{\vec{b}}]^T = E[\vec{x}]\underline{\vec{b}}^T - E[\vec{x}]\underline{\vec{b}}^T = \underline{0}$$

$$\Rightarrow V_x[\underline{A}\vec{x} + \underline{\vec{b}}] = V_x[\underline{A}\vec{x}] + V_x[\underline{\vec{b}}] + \text{Cov}(\vec{x}, \underline{\vec{b}}) + \text{Cov}(\underline{\vec{b}}, \vec{x}) = V_x[\underline{A}\vec{x}]$$

$$\textcircled{2} \quad V_x[\underline{A}\vec{x}] = E_x[(\underline{A}\vec{x})(\underline{A}\vec{x})^T] - E_x[\underline{A}\vec{x}] \cdot E_x[\underline{A}\vec{x}]^T \quad (\text{Def.})$$

$$= E_x[\underline{A}\vec{x}\vec{x}^T\underline{A}^T] - \underline{A}E_x[\vec{x}]E_x[\vec{x}]^T\underline{A}^T$$

$$= \underline{A}E_x[\vec{x}\vec{x}^T]\underline{A}^T - \underline{A}E_x[\vec{x}] \cdot E_x[\vec{x}]^T\underline{A}^T$$

$$= \underline{A} \left(\underbrace{E_x[\vec{x}\vec{x}^T] - E_x[\vec{x}] \cdot E_x[\vec{x}]^T}_{V_x[\vec{x}]} \right) \underline{A}^T = \underline{A}\underline{\Sigma}\underline{A}^T$$

$$(5) \text{Cov}(\vec{x}, \vec{y}) = \underline{\Sigma}\underline{A}^T$$

$$= \text{Cov}(\vec{x}, \underline{A}\vec{x} + \underline{\vec{b}}) = \underline{\Sigma}\underline{A}^T$$

$$\text{Cov}(\vec{x}, \underline{A}\vec{x} + \underline{\vec{b}}) = E[\vec{x}(\underline{A}\vec{x} + \underline{\vec{b}})^T] - E[\vec{x}]E[\underline{A}\vec{x} + \underline{\vec{b}}]^T$$

$$= E[\vec{x}\vec{x}^T\underline{A}^T + \vec{x}\underline{\vec{b}}^T] - \underline{\vec{\mu}} \cdot (\underline{A}\underline{\vec{\mu}} + \underline{\vec{b}})^T$$

$$= E[\vec{x}\vec{x}^T\underline{A}^T] + E[\vec{x}\underline{\vec{b}}^T] - \underline{\vec{\mu}}\underline{\vec{\mu}}^T\underline{A}^T - \underline{\vec{\mu}}\underline{\vec{b}}^T$$

$$= E[\vec{x}\vec{x}^T]\underline{A}^T - \underline{\vec{\mu}}\underline{\vec{\mu}}^T\underline{A}^T - \underline{\vec{\mu}}\underline{\vec{b}}^T$$

$$= E[\vec{x}\vec{x}^T]\underline{A}^T - E_x[\vec{x}]E_x[\vec{x}]^T\underline{A}^T = (E_x[\vec{x}\vec{x}^T] - E_x[\vec{x}] \cdot E_x[\vec{x}]^T)\underline{A}^T$$

$$= \underline{\Sigma}\underline{A}^T$$

9. Statistical Independence

(1) Two random variables X, Y are statistically independent
 $\Leftrightarrow p(\vec{x}, \vec{y}) = p(\vec{x})p(\vec{y}) \checkmark$

★ reverse
NOT true

$\Rightarrow p(\vec{y}|\vec{x}) = p(\vec{y})$ ①
 $\Rightarrow p(\vec{x}|\vec{y}) = p(\vec{x})$ ②
 $\Rightarrow \text{Cov}_{X,Y}(\vec{x}, \vec{y}) = \vec{0}$ ③ $\Rightarrow V_{X,Y}[\vec{x} + \vec{y}] = V_X[\vec{x}] + V_Y[\vec{y}]$

(2) Two random variables X, Y are conditionally independent given Z
 $\Leftrightarrow p(\vec{x}, \vec{y}|\vec{z}) = p(\vec{x}|\vec{z}) \cdot p(\vec{y}|\vec{z})$ for any $z \in Z$

Note: $p(\vec{x}, \vec{y}) = p(\vec{x}|\vec{y}) \cdot p(\vec{y}) \Rightarrow p(\vec{x}, \vec{y}|\vec{z}) = p(\vec{x}|\vec{y}, \vec{z}) \cdot p(\vec{y}|\vec{z})$

Summary:

probability { Basic concepts { Sample space, Event space
Target Space, Probability, Distribution
Sum Rule (marginalization property)
Product Rule $\Rightarrow$ Bayes' Thm
Expectation $\rightarrow$ Mean
Variance $\rightarrow$ Covariance $\rightarrow$ Correlation
Independence. Sec. 6.1 - 6.4

10. Gaussian Distribution.

(1) Univariate: mean μ , Variance σ

probability density function:

$$p(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma}\right)$$

(2) Multivariate: mean $\vec{\mu}$, covariance matrix

$$\Sigma = V_X[\vec{x}] = \text{Cov}_X[\vec{x}, \vec{x}]$$

$$\checkmark p(\vec{x} | \vec{\mu}, \Sigma) = (2\pi)^{-\frac{D}{2}} \cdot \det(\Sigma)^{-\frac{1}{2}} \exp(-\frac{1}{2}(\vec{x}-\vec{\mu})^T \Sigma^{-1}(\vec{x}-\vec{\mu}))$$

for $\vec{x} \in \mathbb{R}^D$ $p(\vec{x}) = \mathcal{N}(\vec{x} | \vec{\mu}, \Sigma)$ or $X \sim \mathcal{N}(\vec{\mu}, \Sigma)$.

(3) Marginal and Conditionals

Let X, Y be two multivariate Gaussian variables

① Concatenate $[\vec{x}^T, \vec{y}^T]^T$ (concatenation is Gaussian)

$$p(\vec{x}, \vec{y}) = \mathcal{N}\left(\begin{bmatrix} \vec{\mu}_x \\ \vec{\mu}_y \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{bmatrix}\right)$$

where $\Sigma_{xx} = \text{Cov}[\vec{x}, \vec{x}]$, $\Sigma_{yy} = \text{Cov}[\vec{y}, \vec{y}]$, $\Sigma_{xy} = \text{Cov}[\vec{x}, \vec{y}]$

② Conditional distribution $p(\vec{x}, \vec{y})$ is also Gaussian.

$$p(\vec{x} | \vec{y}) = \mathcal{N}(\vec{\mu}_{x|y}, \Sigma_{x|y})$$

where $\vec{\mu}_{x|y} = \vec{\mu}_x + \Sigma_{xy} \Sigma_{yy}^{-1}(\vec{y} - \vec{\mu}_y)$, $\Sigma_{x|y} = \Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}$

③ Marginal is still gaussian distribution

$$p(\vec{x}) = \int p(\vec{x}, \vec{y}) d\vec{y} = \mathcal{N}(\vec{x} | \vec{\mu}_x, \Sigma_{xx})$$

$\Rightarrow$ See Example 6.6

(4) Sum and Linear Transformation of Gaussian.

① Sum: X, Y : independent Gaussian random variables

$$X \sim \mathcal{N}(\vec{\mu}_x, \Sigma_x) \quad Y \sim \mathcal{N}(\vec{\mu}_y, \Sigma_y)$$

$$\underline{X+Y} \sim \mathcal{N}(\vec{\mu}_x + \vec{\mu}_y, \Sigma_x + \Sigma_y)$$

② Linear Combination:

a) univariate: $p(ax+by) = \mathcal{N}(a\mu_x + b\mu_y, a^2\Sigma_x + b^2\Sigma_y)$

b) Multivariate: $X \sim \mathcal{N}(\vec{\mu}, \Sigma) \quad \vec{y} = A\vec{x}$

$$Y \sim N(A\vec{\mu}, A\Sigma A^T)$$

Application: obtaining samples from a multivariate $N(\vec{\mu}, \Sigma)$ leveraging a sampler $N(\vec{0}, I)$

Step 1: get a sample from $\tilde{X} \sim N(\vec{0}, I)$

Step 2: compute $\vec{y} = A\tilde{X} + \vec{\mu}$, with $A A^T = \Sigma$ ^{positive definite}
 $\uparrow$
 decompose Σ using
 e.g. Cholesky decomposition

③ Mixture of two univariate Gaussian densities function,
 $p_1(x) = N(\mu_1, \sigma_1)$, $p_2(x) = N(\mu_2, \sigma_2)$, $0 < \alpha < 1$

consider $p(x) = \alpha p_1(x) + (1-\alpha) p_2(x)$

Mean: $\checkmark E[X] = \alpha \mu_1 + (1-\alpha) \mu_2$

$$\checkmark V[X] = [\alpha \sigma_1 + (1-\alpha) \sigma_2] + \left([\alpha \mu_1^2 + (1-\alpha) \mu_2^2] - [\alpha \mu_1 + (1-\alpha) \mu_2]^2 \right)$$

Proof: $V_X[X] = E_X[X^2] - (E_X[X])^2$

$$\begin{aligned} \textcircled{1} \underline{E_X[X]} &= \int_{-\infty}^{+\infty} x p(x) dx = \int_{-\infty}^{+\infty} (\alpha x p_1(x) + (1-\alpha) x p_2(x)) dx \\ &= \underbrace{\alpha \int_{-\infty}^{+\infty} x p_1(x) dx}_{\mu_1} + \underbrace{(1-\alpha) \int_{-\infty}^{+\infty} x p_2(x) dx}_{\mu_2} = \alpha \mu_1 + (1-\alpha) \mu_2 \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \textcircled{2} E_X[X^2] &= \int_{-\infty}^{+\infty} x^2 p(x) dx = \int_{-\infty}^{+\infty} (\alpha x^2 p_1(x) + (1-\alpha) x^2 p_2(x)) dx \\ &= \alpha \int_{-\infty}^{+\infty} x^2 p_1(x) dx + (1-\alpha) \int_{-\infty}^{+\infty} x^2 p_2(x) dx \\ &= \alpha (\mu_1^2 + \sigma_1) + (1-\alpha) (\mu_2^2 + \sigma_2) \quad \textcircled{2} \end{aligned}$$

$$V_X[X] = E_X[X^2] - (E_X[X])^2$$

$$= \alpha(\mu_1^2 + \sigma_1) + (1-\alpha)(\mu_2^2 + \sigma_2) - \underbrace{(\alpha\mu_1 + (1-\alpha)\mu_2)^2}$$

Section 6.5. $= \alpha\sigma_1 + (1-\alpha)\sigma_2 + (\alpha\mu_1^2 + (1-\alpha)\mu_2^2 - [\alpha\mu_1 + (1-\alpha)\mu_2]^2)$

More different distributions Section 6.6. self-study.

How to transform known distribution to another new distribution? $\rightarrow$ Section 6.7 self-study.